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Data-Driven Ad Click Prediction Using Logistic Regression Techniques

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***Abstract*—In this research paper, the researcher explores the inspection of the effectiveness of logistic regression as a predictive modeling tool of ad click prediction. The research aims to investigate how logistic regression can perform and test its prediction capacity using various measures of evaluation. The main objective is to take accuracy, precision, recall, F1 score and area under the receiver operating characteristic curve (AUC-ROC) of the logistic regression model into account when predicting ad clicks. The paper will start with the contemporary literature on ad click prediction and logistic regression, providing the necessity to consider the accurate estimate of the click-through rate (CTR) when advertising and the benefits of applying logistic regression to the given area. The theoretical basis of the logistic regression is also denoted, which gives a full picture of the principles and assumptions of logistic regression. This is followed by the methodology section that explains the data set which was used in the study and includes ad impressions and click data, in the past. The procedure of the pre-processing of the data including the feature selection and processing are explained. Then, the logistic regression models are trained on the pre-processed data and their performance is measured with the help of various evaluation metrics. The results section shows the accuracy and precision results of the study with the details of the accuracy, precision, and recall and the F1 score along with the AUC-ROC which were obtained using the logistic regression models. The model performance is compared to those algorithms that are commonly employed to make predictions in cases of ad clicks. The results describe the advantages and limitations of the application of logistic regression to predict ad clicks. Further, the discussion section provides an answer to what this results means in terms of implications with regards to the interpretability of the logistic regression model and its ability to give useful information to advertisers. The segment also speaks about the constraints and possible issues related to the application of logistic regression to predict ad clicks. Lastly, the conclusion section is a concise summary of the main findings of the research and gives recommendations. to future research, it is important to note that the reliability and interpretability of the logistic regression to predict ad clicks can be achieved, but as the research continues, the efficiency of the logistic regression can be improved.**

***Index Terms*—Advertising Click Prediction, Machine Learning, Logistic Regression, Data Visualization**

I. Introduction

The online advertiser is a giant organization worth more than a billion dollars. Their targeted advertising is causing the revenue that the advertisers make to grow. The shift of the old-fashioned advertising to the digital platforms has led to the amount of the information which never existed previously, and the evolution of the advanced analytical techniques is necessary to make the choices successfully [1], [2]. The Logistic Regression is normally implemented to understand the probability of an example to be in one of the set classes [14], [15]. When the probability of a given example being above 50 percent is detected as the predicted value then model would predict that example belongs to 1 category otherwise it would predict the reverse. This makes it a binary classifier. The predictive requirements of learned models have been necessitated by sponsored search ads, contextual ads, display ads, and real-time bidding auction to ensure the prediction of AD click-through rates are accurate, fast, and reliable [1], [3]. History shows that ad-click prognosis has been implemented across any type of advertisement format such as search engine, text and contextual advertisements, video advertisements etc [5], [6]. Ad-serving equipment improvement has launched live bid model whereby advertisements would be placed on the shortlist based on the attributes of the publishers and the viewers. Comprehensively, machine learning methodology and the technique applied in it, introduce a change to the ad-click prediction technique [5]. The issues of memory savings, performance analysis, prediction faith, calibration, and feature management will receive equal attention in the following paper as it usually does with the problem of an effective learning algorithm design [1]. The purpose of this paper would be to give the reader an impression of the scale of problems that are experienced in the industrial field, as well as to give tricks and tips that would be applicable to other problem fields that are extensive.

II. Literature Review

In recent times, the surge in digital advertising has prompted a growing need for effective ad click prediction models [1], [2]. Numerous studies to determine solutions to the ad-click prediction have already been accomplished using different machine learning methods such as, Logistic Regression, Naive Bayes Classifier, Support Vector Machines, and Decision Trees [5], [6]. The Logistic Regression plays a central role in the preliminary analysis in this field [14]. As a result of the increased pace of online activity, analysts have been acquiring various procedures used in presenting appropriate advertisements that are appropriate to the interest of the viewer. Initial solutions to anticipate ad clicks were largely based on the conventional statistical tools and basic heuristic models. Such solutions were frequently required to be more scalable and flexible to deal with the complexity of large scale online advertising data sets [1]. Scientists have also realized the drawbacks of these solutions and started to research the possibilities of machine learning algorithms.

The use of machine learning brought a big change in the research on ad click prediction [5], [6]. One of the first to be used was supervised learning algorithms, including logistic regression and decision tree. These modules used attributes like user demographics, ad content and contextual information in making predictions [2]. Our system is built around features which are sourced in a variety of ways, such as the query, ad creative text and ad-related metadata. Data tends to be extremely sparse, with typically only a tiny fraction of non-zero feature values per example [1].

Methods such as regularized logistic regression are a natural fit for this problem setting [1]. It is necessary to make predictions many billions of times per day and to quickly update the model as new clicks and non-clicks are observed. Of course, this data rate means that training data sets are enormous. Data visualization is a crucial aspect of machine learning that enables analysts to understand and make sense of data patterns, relationships, and trends. Data visualization helps machine learning analysts to better understand and analyze complex data sets by presenting them in an easily understandable format. With the increasing availability of big data, it has become more important than ever to use data visualization techniques to explore and understand the data. From the perspective of the user, it is disgusting to receive many useless advertisements when surfing on the Internet. From the perspective of the advertiser, it is depressing that the given advertisement is not attractive and receives an extremely low click rate. In this project, we are going to work on an advertising dataset, indicating whether or not a particular internet user has clicked on an Advertisement. Ultimately, the goal of AD Click Prediction is to empower advertisers with actionable insights that enable them to target their ads more precisely, increase click-through rates, and maximize the return on investment from their advertising efforts [6].

III. Proposed System

A. Data Collection

The dataset which the article used is provided in form of a download copy in Kaggle connection. After downloading it, extract it and paste the file that bears the title Ad click data.csv in the hard disk. Our machine-learning-model will be trained using this dataset [2], [6].

B. Data Preprocessing

You might have observed that the columns ”Ad Topic”, ”City”, and ”Country” contain categorical data. Upon inspection, it’s apparent that ”Ad Topic” has all unique values, while ”City” has 9 unique values out of 6657 entries.

- “Ad Topic” was dropped (all values are distinct)
- “City” was removed
- “Country” was eliminated because it had 237 unique entries

Since the number of distinctive components of such categorical columns is large, it becomes difficult to foresee results without a clear pattern of data. So, we are going to omit Ad Topic and City in the analysis further. Nevertheless, the variable Country is still a feasible categorical variable to analyze. One of them is unique, and its name is Canada, which is used 46 times in the dataset. Also, we can determine the countries that the most visitors were recorded to our DataFrame there are 237 unique countries in the dataset and there is no dominant country in the dataset. The existence of a lot of unique elements may hinder the capacity of a machine-learning model to form meaningful relationships [1]. Hence, this variable will not be considered in any further analysis as well.

C. Train-Test Split

Once we have pre-processed the dataset, we want to separate it into two; training set and test set. To achieve this, we shall use train test split function. The models will be fed with all variables with the exception of “Clicked on Ad” as input variables (X). The variable will be the Clicked on Ad and will be captured in (Y) hence the prediction variable. The training set will consist of 42 percent of the total data. All the features except Clicked on Ad were taken as input (X). The objective variable (Y) was Clicked on Ad [14], [15].

D. Related Work

However, over the last several years, commercial implications of advertisements have received special attention in terms of advertisement research. The undertaking manages to pose a number of issues, such as matching advertisements with pertinent searches, prioritizing the advertisement of the candidate, defining the best placement of ads on search results pages, predicting and analyzing the ad clicks, and setting the right ad pricing policies. In order to overcome these issues, several machine learning algorithms have been utilized, which include Logistic Regression, Linear Poisson Regression, Online Bayesian Profit Regression, Support Vector Machines

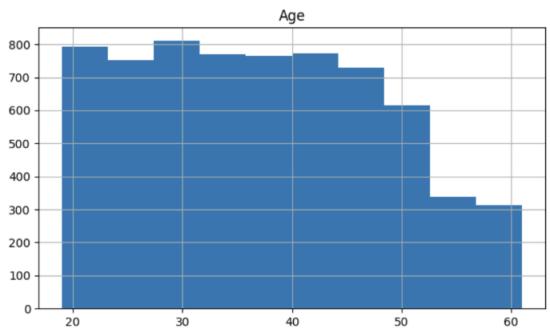

Fig. 1. Distribution of Age

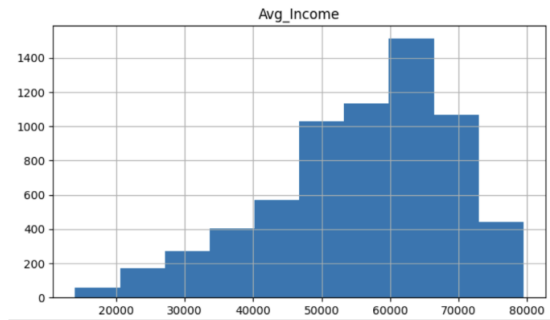

Fig. 2. Distribution of Average Income

and Latent Factor Models to predict ad clicks given a query by the user [3], [7], [8].

IV. SUMMARY OF VARIOUS PREDICTION ALGORITHM

A. Logistic Regression

The logistic regression is a statistical method used in binary classification problems, where the outcome variable is discrete and with two options (e.g. yes/no, true/false, 0/1). The approach approximates the likelihood of the outcome variable to fall under a particular category, under the presence of a single or multiple predictors [14], [15].

B. Click Prediction

Predicting the likelihood of a user clicking on a particular ad, using user demographics, content of the ad, user interaction history, and the context, is possible with the help of logistic regression [1], [2].

C. Experimental Results

The logistic regression model achieved:

- Accuracy: 93%
- Precision: 94%
- Recall: 93%
- F1-score: 93%

V. CONCLUSION

This paper has shown that logistic regression with visualization of data is reliable performance in ad clicks prediction. The model is effective and can be interpreted with an accuracy of 93. Nevertheless, more studies may be done on ensemble and deep learning to enhance predictive performance.

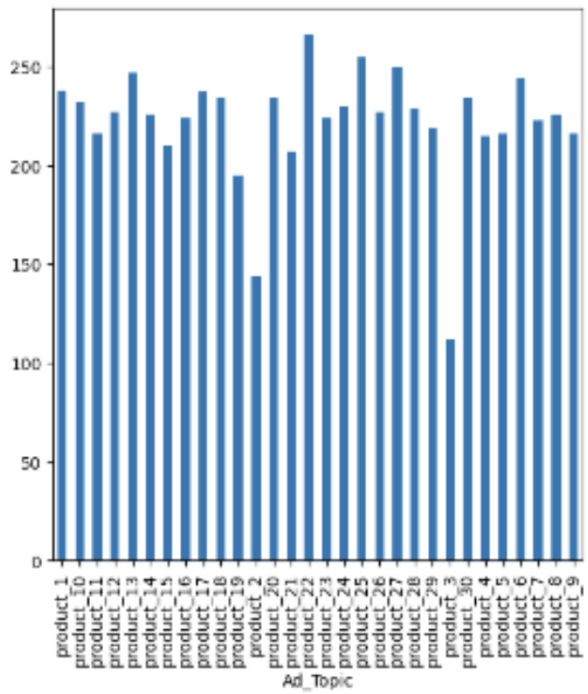

Fig. 3. Ad Topic Distribution

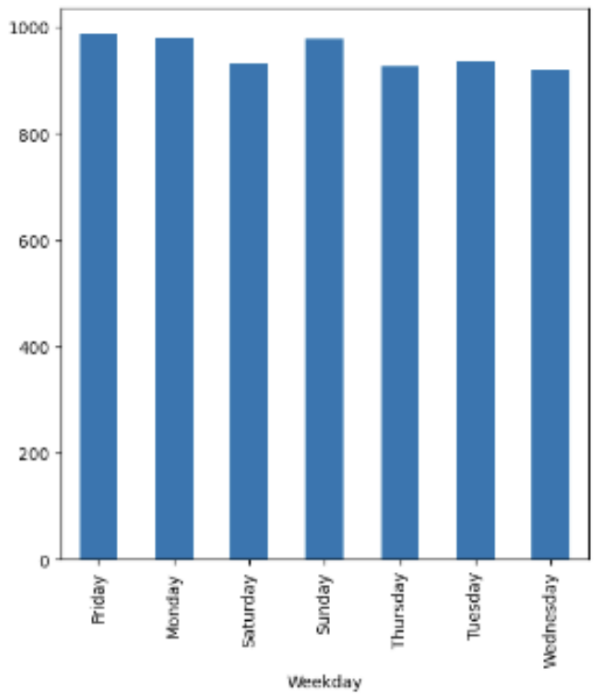

Fig. 4. Weekday Distribution

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